Full-Stack Engineering Prompt & Stage-by-Stage Implementation Blueprint for Antigravity AI: 2026 Tesla Roadster Flying Car Simulation, 1,048-Part 3D CAD Model, LLM Autonomy & 10 Adventure Proving Grounds

Master Multi-Stage Execution Specification for Building the Photorealistic 3D CAD Assembly, Coupled Real-Gas COPV & Underbody Suction Physics Engine, Autonomous VLA/MPC Flight-Track Brain, and 10 Interactive Proving Ground Trails

This specification provides the complete, copy-pasteable **Stage-by-Stage Full-Stack Engineering Prompt** designed for execution inside **Antigravity AI** (and compatible agentic IDE/simulation pipelines alongside Unreal Engine 5, MATLAB/Simulink, Assetto Corsa, and BeamNG.drive). Each stage is self-contained, deterministic, and grounded in the verified October 15–16, 2026 Tesla Roadster prototype specifications revealed at SpaceX McGregor, Texas (CellCog; EVXL; Kingy AI; Paul Tan).

Execution Stage	Target Subsystem & Deliverable Module	Primary Tech Stack & Solvers	Verification Gate Criteria
Stage 0: Master System Persona & Global Constraints	Chief Engineer System Prompt, Physical Constants, & Three-Trim Configurator (Performance, Thrust, Rocket)	TypeScript 5.6 + Python 3.12 + WebGPU / Three.js r170 + FastAPI	Zero placeholder values; exact match to Oct 2026 McGregor telemetry
Stage 1: Monorepo Architecture & 1 kHz Telemetry Bus	Full-Stack Project Scaffolding, Shared Binary State Ring-Buffer, & WebSocket/gRPC Bridge	Vite + React 19 + Three.js + Rapier3D WASM + Python Asyncio	≥120 FPS 3D render loop decoupled from deterministic 1,000 Hz physics tick
Stage 2: 1,048-Part Parametric 3D CAD Scene Graph	Hierarchical 10-Subsystem CAD Builder (`RD26-0001` to `RD26-1048`), PBR Materials, & Exploded-View Kinematics	Procedural CAD CSG/NURBS + InstancedMesh + MediaPipe Hand Gestures	All 1,048 parts selectable, inspectable, and animated with kinematic joints
Stage 3: Coupled 6-DOF Rocket, Suction & EV Physics	Real-Gas COPV Blowdown ($Z=1.192$), 10 Supersonic CGT Nozzles, 4 × 20 kW Suction Fans, & 1 MW RWD	RK4 6-DOF Rigid-Body + CoolProp Lookup LUT + Pacejka Tire Model	Reproduces 0–60 mph in 1.55s (Perf), 0.81s (Thrust), 0.55–0.59s (Rocket)
Stage 4: Autonomous LLM/VLA Brain & 1 kHz SpaceX GNC	Tier-1 Multi-Modal VLA Strategic Planner + Tier-2 250 Hz Convex MPC + Tier-3 1 kHz Control Allocator	OpenAI/Gemini/Grok VLA Schema + OSQP Convex Solver + Quaternion LQR	Zero-intervention execution of inverted wall climb, kickflip, & hover-slam
Stage 5: 10 Interactive 3D Adventure Proving Grounds	Procedural Terrain Meshes, Atmospheric Density Profiles, Friction Maps, & Mission State Machines	WebGPU Terrain Shaders + Niagara/GPUParticles Mach Diamond Plumes	All 10 adventures completable autonomously with live telemetry scoring

Execution Stage	Target Subsystem & Deliverable Module	Primary Tech Stack & Solvers	Verification Gate Criteria
Stage 6: Track Mode HMI, Industry Exports & Test Suite	Ultrawhite/Red Cockpit Telemetry HUD + MATLAB, Assetto Corsa, BeamNG, & Unreal Engine 5 Exporters	React Canvas HUD + Automated Pytest/Vitest Physics Validation Suite	100% pass across 25 physical assertion tests and 4 external sim exporters

Usage Protocol for Antigravity AI: Paste **Stage 0** first as the persistent system/project context, then feed **Stages 1 through 6** sequentially. Wait for Antigravity AI to generate, execute, and verify all files in each stage before advancing to the next stage.

Stage 0: Master System Persona, Architectural Directives & Physical Constants Prompt

Copy and paste the following prompt block into Antigravity AI as the foundational project specification and persistent system prompt:

```
[SYSTEM ROLE & CHIEF ENGINEER DIRECTIVE FOR ANTIGRAVITY AI]
Act simultaneously as:
1. Chief Vehicle & Propulsion Architect at Tesla Motors and SpaceX (McGregor/Hawthorne/Giga
Texas).
2. Principal 6-DOF Flight Control, Aerothermodynamics, and Autonomous Vision-Language-
Action (VLA) Systems Engineer.
3. Lead Full-Stack 3D Graphics, WebGPU/Three.js, CAD Kinematics, and Physics Simulation
Architect.

MISSION OBJECTIVE:
Build a complete, production-grade, interactive 3D Engineering Simulation, Parametric
1,048-Part CAD Explorer, Autonomous LLM/VLA Control System, and 10-Adventure Proving Ground
Application for the 2026 Next-Generation Tesla Roadster (covering all three official
October 15, 2026 variants: Performance, Thrust Augment, and Rocket Augment).

STRICT ENGINEERING & IMPLEMENTATION RULES:
1. ZERO MOCKS OR TOY PLACEHOLDERS: Every physical equation (isentropic & real-gas Z=1.192
COPV blowdown, choked De Laval supersonic nozzle thrust with ambient backpressure,
underbody suction fan pressure-leakage FSI, Pacejka nonlinear tire friction, and 6-DOF
quaternion rigid-body dynamics) must be implemented with exact mathematical equations.
2. COMPLETE 1,048-PART HIERARCHICAL 3D CAD SCENE GRAPH: Construct the full 10-subsystem
parametric CAD assembly (`RD26-0001` through `RD26-1048`) with individual part metadata,
materials (Toray T1100G CFRP, Ti-6Al-4V, Inconel 718, Mg-AZ91D, Al-Li 2195, Gold Glazed
TiO2 glass, Ano White coating), real-time exploded view, subsystem isolation, and kinematic
articulation.
3. VERIFIED OCTOBER 2026 REAL SPECIFICATIONS:
   - Variants:
     * Performance (2 seats, ~1,700 kg, 0-60 mph <1.60s, 4x20 kW suction fans + active
aero, $50k reservation).
     * Thrust Augment (2 seats, ~1,900 kg loaded, 0-60 mph <0.90s / 0.85s / 0.81s live
demo, adds 4 rear diffuser cold-gas thrusters >80 kN + 200L trunk COPV, $250k reservation).
     * Rocket Augment (1 seat single-seater, ~2,080 kg dry / ~2,350 kg wet, 0-60 mph <0.59s
/ 0.55s rollout @ 4.64g avg, 10 perimeter cold-gas thrusters [4 rear main, 4 wheel-well
downward lift/roll, 2 nose lateral yaw] + 600L total COPV [200L trunk + 400L passenger
cabin storing 227-250 kg N2/air at 5,500 psi / 379.2 bar working pressure and 10,000 psi /
689.5 bar vessel rating], invite-only).
   - Electric Powertrain: Rear-Wheel Drive (RWD) only, 2 rear motors (>1.0 MW / 1,341-1,500
```

```
hp total, >1,400 Nm torque), no differential, dual independent SiC inverters, twin 8.4:1
planetary gearsets, 800-1,000V aluminum-can 4680 battery with double-sided cross-flow
cooling and tailor-welded titanium upper plate, 500 kW V4 Supercharging (15-80% in 10 min).
   - Ground-Effect Suction System (US Patent 12,377,920): 4 x 20 kW rear diffuser fans (80
kW total, 28,000 RPM) + 4 independent deployable rubber/nylon-bristle sealing skirts across
a 4.0 m^2 floor plenum generating 11.52 kPa vacuum (4,700 kgf / 46.1 kN / 2.0g downforce at
0 mph), with 20.8 kW/mm skirt gap leakage sensitivity and automatic transition above 200
mph to High-Speed Open-Skirt Venturi Mode.
   - Active Aerodynamics: Optimus humanoid robot leg/ankle actuators driving the rear wing
(0-90 deg pitch, 700 kg passive downforce, 90 deg airbrake), 5-surface front wing (up to 80
deg tilt), pop-out front brake cooling barge boards, and hood vent.
   - Cockpit & Controls: Fixed chassis-molded carbon seat buckets, motorized movable pedal
box, adjustable steer-by-wire yoke (+/-170 deg travel), 4-wheel independent by-wire carbon-
ceramic brakes, Track Mode adaptive spool-valve suspension (-40 mm drop, 3x stiffness).
```

Stage 1: Full-Stack Monorepo Scaffolding, State Architecture & 1 kHz Telemetry Bus

Feed the following Stage 1 prompt to Antigravity AI to initialize the repository structure, shared state schemas, and deterministic physics-to-render synchronization pipeline:

```
[STAGE 1 PROMPT — EXECUTE IN ANTIGRAVITY AI]
Initialize the full-stack monorepo `tesla-roadster-2026-sim/` with the following
architecture and generate all core configuration, schema, and state-bus files:

1. DIRECTORY ARCHITECTURE:
   - `packages/shared-Core/`: TypeScript & Python mirrored data contracts for vehicle
state, 1,048-part BOM metadata, COPV thermodynamics, 10 CGT nozzles, 4 suction fans, VLA
commands, and adventure telemetry.
   - `apps/web-simulator/`: Vite + React 19 + TypeScript + Three.js (WebGPU/WebGL2) +
`@react-three/fiber` + `@react-three/drei` + TailwindCSS + MediaPipe Hands (21-point
gesture CAD controller).
   - `services/physics-autonomy-engine/`: Python 3.12 + FastAPI + WebSockets + NumPy/SciPy
+ CoolProp/Virial real-gas module + OSQP 250 Hz Convex MPC + LLM/VLA Strategic Planner
agent.
   - `exporters/`: Automated bridge generators for MATLAB/Simulink (`.m`/`.slx` params),
Assetto Corsa (`car.ini`, `aero.ini`, `drivetrain.ini`, `suspensions.ini`,
`thrusters.lua`), BeamNG.drive (`roadster26.jbeam`, `suction_cgt.lua`), and Unreal Engine
5.5 (`ChaosRoadsterConfig.json` + C++ header).

2. CORE STATE VECTOR (`VehicleState6DOF`):
   Define a strictly typed 64-float contiguous `Float64Array` / `SharedArrayBuffer` state
vector updated at 1,000 Hz (dt = 0.001 s):
   - Inertial Position: `pos = [x, y, z]` (m) and Velocity: `vel = [vx, vy, vz]` (m/s)
   - Body Attitude Quaternion: `quat = [qw, qx, qy, qz]` and Body Angular Velocity: `omega
= [wx, wy, wz]` (rad/s)
   - Active Variant Enum: `PERFORMANCE` | `THRUST` | `ROCKET`
   - COPV Thermodynamic State: `copvVolumeL` (0 | 200 | 600), `gasMassKg` (0..250.0),
`stagnationPressureBar` (1.013..689.5, nominal 379.2), `stagnationTempK` (140.0..320.0,
nominal 293.15), `compressibilityZ` (1.0..1.49), `compressorActive` (boolean)
   - 10 Cold-Gas Thruster States (`cgt[0..9]`): `valveOpenRatio` (0.0..1.0 driven by
Optimus actuator dynamics with tau = 1.2 ms), `gimbalPitchRad` (-0.14..+0.14),
`gimbalYawRad` (-0.14..+0.14), `massFlowKgS`, `exitMach`, `instantThrustN`
   - Underbody Suction State: `fanRpm[0..3]` (0..28,000 RPM), `fanPowerKw` (0..80.0),
`skirtGapMm[0..3]` (1.5..55.0 mm), `plenumVacuumKpa` (0..13.5 kPa), `suctionDownforceN`
(0..46,100 N), `venturiBypassOpen` (boolean, true when speed > 89.4 m/s [200 mph])
   - Active Aero State: `rearWingPitchDeg` (0..90 deg), `frontWingFlapDeg` (0..80 deg),
`bargeBoardDeployMm` (0..65 mm), `hoodVentOpenRatio` (0..1), `passiveAeroDownforceN`,
```

```
`aeroDragN`
   - RWD Electric Powertrain & Chassis State: `motorTorqueNm[2]` (Left/Right rear motors,
0..850 Nm each at shaft, 8.4:1 planetary ratio), `inverterPowerKw[2]` (0..550 kW each),
`batterySocPct` (0..100%), `batteryVoltageV` (800..1000 V), `batteryTempC`,
`wheelSlipRatio[4]`, `wheelSlipAngleRad[4]`, `tireNormalLoadN[4]`,
`suspensionDeflectionMm[4]`, `rideHeightOffsetMm` (-40..+35 mm), `yokeAngleDeg` (-170..+170
deg), `pedalBoxOffsetMm` (0..220 mm).

Write the complete `packages/shared-core/src/vehicleState.ts` and `services/physics-
autonomy-engine/state_schema.py` files now.
```

Stage 2: Parametric 1,048-Part 3D CAD Scene Graph, Aerospace Shaders & Exploded-View Kinematics

Once Stage 1 compiles, issue the Stage 2 prompt to generate the procedural 1,048-part 3D CAD assembly, materials library, and touchless MediaPipe hand-gesture CAD inspection interface:

```
[STAGE 2 PROMPT — EXECUTE IN ANTIGRAVITY AI]
Build the complete 1,048-part hierarchical 3D CAD model and kinematic scene graph in
`apps/web-simulator/src/cad/` covering all 10 engineering subsystems (`SYS-01` through
`SYS-10`, Part IDs `RD26-0001` to `RD26-1048`):

1. PARAMETRIC CAD GENERATOR (`RoadsterCADAssembly.ts`):
   Generate a deterministic scene graph containing all 1,048 parts grouped into 10 top-
level `THREE.Group` subsystem nodes and 28 sub-assembly nodes:
   - `SYS-01` (`RD26-0001`..`0112`, 112 parts, 245.0 kg): Carbon-Ti monocoque tub with
integrated fixed seat shells, front Mg-AZ91D double-wishbone subframe, rear Ti-6Al-4V
electron-beam-welded thrust-structure subframe, tubular A/B-pillar roll halo, and spherical
COPV cradle bulkheads.
   - `SYS-02` (`RD26-0113`..`0210`, 98 parts, 138.0 kg): SR-71-inspired sharp wedge body
panels in `Ano White` (custom PBR shader: diffuse `#F8FAFC`, clearcoat 1.0, roughness 0.12,
subtle cryogenic pearl fresnel), upward-opening dihedral doors (`revolute` hinge: 68 deg
upward, 22 deg outward), `Gold Glazed` spacesuit-visor glass (transmission 0.72, specular
tint `#D97706`, IOR 1.52, 68.5 deg windshield rake), 2 removable T-top roof panels
(storable in rear trunk), 15 mm pop-down micro-LED headlights, full-width front/rear
lightbars, and removable cowl cap with clip-in windshield wiper.
   - `SYS-03` (`RD26-0211`..`0304`, 94 parts, 64.5 kg): Active rear wing on single central
pylon + 4 Optimus robot ankle actuators (2 exposed, 2 concealed) with mechanical friction
brake (0-90 deg airbrake articulation) + hidden left-rear NACS charge port door; 5-surface
front wing with 80 deg tilting leading edges; pop-out front brake barge boards (0-65 mm
lateral slide); active hood radiator louver.
   - `SYS-04` (`RD26-0305`..`0412`, 108 parts, 78.5 kg): 4 x 20 kW rear diffuser suction
fans (220 mm Ti-6Al-4V mixed-flow blisks spinning up to 28,000 RPM) mounted directly below
the rear thruster bezel; 4 independent drop-down Viton rubber & Nylon-66 bristle sealing
skirts (`prismatic` Z-travel 0-55 mm); 4.0 m^2 Venturi plenum floor with 200 mph bypass
gates.
   - `SYS-05` (`RD26-0413`..`0556`, 144 parts, 112.0 kg): 10 SpaceX Cold-Gas Thruster (CGT)
supersonic De Laval nozzles:
     * `CGT-R1..R4`: 4 gold-lined Inconel 718 rear diffuser main nozzles (d_throat = 21.4
mm, expansion ratio 2.05:1, +/-8 deg 2-axis gimbal) pointing aft (-X).
     * `CGT-W1..W4`: 4 Ti-6Al-4V downward-pointing lift/roll nozzles mounted behind each
wheel arch pointing normal to the chassis floor (-Z).
     * `CGT-N1..N2`: 2 flush-scarfed Ti-6Al-4V lateral yaw nozzles mounted in the
left/right front nose fenders pointing (+/-Y).
   - `SYS-06` (`RD26-0557`..`0678`, 122 parts, 195.0 kg dry): 200-liter trunk COPV + 400-
liter passenger-bay COPV (filament-wound Toray T1100G carbon overwrap mesh over Al-Li 2195
liner), 38 mm Inconel 718 / MP35N high-pressure feed manifolds, 10 Optimus high-speed
pintle valves, 750 bar burst discs, and 4-stage intercooled compressor + zeolite desiccant
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```
dryer.
   - `SYS-07` (`RD26-0679`..`0794`, 116 parts, 445.0 kg): 800-1000V structural battery pack with aluminum-can 4680 cells, single-pass cross-flow cooling manifolds, tailor-welded Ti-6Al-4V top plate, Al-6061 lower skid plate, and shared Octovalve heat pump.
   - `SYS-08` (`RD26-0795`..`0892`, 98 parts, 124.0 kg): Coaxial 1 MW dual-motor Rear Drive Unit (500 kW per motor, FeCoV stator laminations), dual SiC inverters, twin 8.4:1 planetary reduction gearsets (no differential), and titanium half-shafts.
   - `SYS-09` (`RD26-0893`..`0984`, 92 parts, 186.0 kg): Pushrod double-wishbone front suspension with inboard rocker arms, 5-link rear suspension, gold-anodized spool-valve adaptive dampers, carbon-ceramic rotors with 4 independent by-wire calipers, ultra-thin 6-spoke forged magnesium wheels (20" front, 21" rear), and Pirelli P Zero Trofeo RS "Elect" tires (295/30R20 front, 335/30R21 rear).
   - `SYS-10` (`RD26-0985`..`1048`, 64 parts, 62.0 kg): Steer-by-wire microsuede yoke (+/-170 deg) with boost/turn paddles, motorized sliding pedal box (220 mm travel), driver-angled center touchscreen + 2 forward displays + thin windshield-base strip display, Starman test mannequin toggle, 8 FSD cameras, and triple-redundant 2 kHz SpaceX IMU.

2. SUPERSONIC MACH DIAMOND PLUME & SUCTION VORTEX SHADERS (`ThrusterPlumeShader.ts`):
   Create a custom volumetric raymarched / GPU particle shader attached to each of the 10 CGT nozzle exit planes that renders realistic supersonic condensation plumes with periodic Prandtl-Meyer expansion fans and luminous standing oblique shock waves (Mach diamonds) scaled dynamically by `instantThrustN` and `exitMach`. Add underbody airflow streamlines visualizing the 4 suction fans evacuating the floor plenum.

3. EXPLODED-VIEW KINEMATICS & TOUCHLESS GESTURE CAD CONTROLLER (`GestureCADController.ts`):
   Implement a continuous radial/axial exploded-view displacement field (`explosionFactor` 0.0 to 1.0) that separates all 10 subsystems and 1,048 parts along their assembly normals, plus an optional webcam MediaPipe 21-point 3D hand-landmark controller (Two-hand pinch = FOV zoom; Closed-fist quaternion rotation = 3D orbit; Open-palm translation = pan; Two-fist pull-apart = real-time exploded-view separation with Kalman filter smoothing).
```

Stage 3: First-Principles 6-DOF Physics Engine (COPV Real-Gas Blowdown, 10 CGTs & 4 Suction Fans)

Feed Stage 3 to Antigravity AI to implement the coupled aerothermodynamic, pneumatic, ground-effect, and multi-body vehicle dynamics solver:

```
[STAGE 3 PROMPT — EXECUTE IN ANTIGRAVITY AI]
Implement the deterministic 1,000 Hz (dt = 0.001 s) 4th-Order Runge-Kutta (RK4) 6-DOF physics engine in `services/physics-autonomy-engine/physics_solver.py` and mirror its fast WASM/TypeScript solver in `apps/web-simulator/src/physics/RoadsterPhysicsEngine.ts` using these exact governing equations:

1. REAL-GAS (Z = 1.192) ADIABATIC / POLYTROPIC COPV BLOWDOWN SOLVER:
   - Total COPV volume: `V_tank = 0.20 m^3` (Thrust) or `0.60 m^3` (Rocket). Initial state at full charge: `P_0(0) = 379.2e5 Pa` (5,500 psi, or configurable up to `689.5e5 Pa` / 10,000 psi burst-rated test mode), `T_0(0) = 293.15 K`, real-gas compressibility `Z(P,T) = 1.0 + 0.192 * (P / 379.2e5)`, initial gas mass `m_gas(0) = 227.0 to 250.0 kg`.
   - At each timestep `dt`, sum the mass flow rates across all 10 nozzles:
     `m_dot_total = sum_{k=0..9} m_dot_k`
   - Update remaining gas mass: `m_gas(t + dt) = max(m_residual, m_gas(t) - m_dot_total * dt)`
   - Polytropic/adiabatic blowdown with carbon-wall heat transfer coefficient `h_wall`:
     `T_0(t + dt) = T_0(t) * (m_gas(t + dt) / m_gas(t))**(gamma - 1) + (h_wall * A_wall * (T_ambient - T_0(t)) / (m_gas(t) * c_v)) * dt`
     `P_0(t + dt) = Z(P,T) * (m_gas(t + dt) / V_tank) * R_spec * T_0(t + dt)`
   - Enforce minimum sonic choking check: if `P_0(t) <= 1.893 * P_ambient`, transition nozzle equation to subsonic compressible orifice flow.
```

```
2. 10 SUPERSONIC DE LAVAL COLD-GAS THRUSTER NOZZLES:
   - Rear 4 Main Nozzles (`k = 0..3`): throat diameter `d_t = 0.0214 m` (`A_t = 3.597e-4 m^2` each, `14.39 cm^2` total), expansion ratio `epsilon = A_e / A_t = 2.05` (`M_exit = 2.18`), mounting positions at rear diffuser `r_k = [-1.95, y_k, 0.38] m`, nominal thrust direction `n_k = [+1, 0, 0]` (plus +/-8 deg gimbal pitch/yaw).
   - Wheel-Well 4 Lift/Roll Nozzles (`k = 4..7`): throat diameter `d_t = 0.0145 m`, mounting positions behind FL, FR, RL, RR wheels `r_k = [+/-1.30, +/-0.90, 0.32] m`, thrust direction `n_k = [0, 0, +1]` (upward lift force on vehicle body when exhausting downward).
   - Nose 2 Lateral Yaw Nozzles (`k = 8..9`): throat diameter `d_t = 0.0120 m`, mounting positions `r_k = [+1.85, +/-0.88, 0.42] m`, thrust direction `n_k = [0, -/+1, 0]`.
   - Choked mass flow and under-expanded real-gas thrust per nozzle `k`:
     `m_dot_k = u_k(t) * C_d_nozzle * P_0(t) * A_t_k * sqrt(gamma / (R_spec * T_0(t))) * (2 / (gamma + 1))**((gamma + 1) / (2 * (gamma - 1)))`
     `F_cgt_k = m_dot_k * V_eff_sea_level * (P_a_ref / P_ambient(altitude))**0.08`
     where `V_eff_sea_level = 565.0 m/s` (`I_sp = 57.6 s`), yielding `>80,000 N` (`81.8 kN`) across the 4 rear nozzles at `P_0 = 379.2 bar`.

3. UNDERBODY SUCTION FAN & DYNAMIC SKIRT SEALING FSI MODEL:
   - Effective sealed plenum area `A_plenum = 4.00 m^2`, perimeter `L_p = 13.0 m`.
   - For commanded fan shaft power `P_fan` (up to `4 x 20 kW = 80 kW`) and dynamic skirt clearance gap `h_gap` (m) over local terrain roughness:
     Solve steady-state vacuum equilibrium where fan extraction flow equals skirt perimeter orifice leakage:
     `Q_leak = C_d_skirt * L_p * h_gap * sqrt(2 * DeltaP_plenum / rho_air(altitude))`
     `P_fan * eta_fan = Q_leak * DeltaP_plenum  ==>  DeltaP_plenum = [ (P_fan * eta_fan) / (C_d_skirt * L_p * h_gap * sqrt(2 / rho_air)) ]**(2/3)`
   - Clamp `DeltaP_plenum <= 13,200 Pa` (yielding `46,100 N = 4,700 kgf` at `DeltaP = 11.525 kPa` and `h_gap = 3.5 mm`).
   - Above `200 mph` (`89.4 m/s`), automatically open front/rear skirt bypass gates (`venturiBypassOpen = true`) and transition underbody downforce to high-speed Venturi ground-effect law `F_venturi = 0.5 * rho_air * C_L_venturi * A_ref * v_x^2`.

4. ACTIVE AERODYNAMICS, 1 MW DUAL-MOTOR RWD & PACEJKA TIRE MODEL:
   - Compute passive + active downforce and drag from `rearWingPitchDeg` (0..90 deg, generating up to `6,867 N [700 kgf]` downforce at 160 mph and `18.5 kN` airbrake drag at 90 deg) and `frontWingFlapDeg` (0..80 deg).
   - Compute normal load `F_N_i` on each of the 4 tires including gravity projection `m * g * cos(theta_surface)`, centrifugal wall curvature load `m * v^2 / R_curve`, suction downforce `DeltaP_plenum * A_plenum`, passive wing downforce, and pitch/roll load transfer.
   - Apply Pacejka Magic Formula ($mu_peak = 1.26$ on dry asphalt for Pirelli P Zero Trofeo RS Elect `295/30R20` and `335/30R21`) with independent left/right rear motor torque (`500 kW` / `700+ Nm` per motor through `8.4:1` gearset) and 4-wheel independent carbon-ceramic braking.
```

Stage 4: Autonomous Multi-Modal LLM/VLA Brain & 1 kHz SpaceX GNC Flight-Track Controller

Feed Stage 4 to Antigravity AI to build the three-tier autonomous intelligence and flight/track control system:

```
[STAGE 4 PROMPT — EXECUTE IN ANTIGRAVITY AI]
Implement the three-tier autonomous control stack in `services/physics-autonomy-engine/autonomy_stack.py` and `apps/web-simulator/src/autonomy/VLABrainController.ts`:

1. TIER 1 — STRATEGIC MULTI-MODAL LLM / VLA PLANNER (10-25 Hz):
   - Expose a structured LLM/VLA reasoning agent that ingests:
     (a) Downsampled 8-camera FSD occupancy grid & forward terrain heightmap/curvature
```

```
profile (50 m horizon),
     (b) Real-time telemetry summary (`speedMph`, `pitchDeg`, `rollDeg`, `altitudeM`,
`gasMassKg`, `copvPressurePsi`, `suctionDownforceKgf`, `surfaceMu`),
     (c) Active Adventure Mission Objective (`ADV-01` through `ADV-10`).
   - Output a deterministic JSON Tactical Plan (`VLATacticalCommand`):
     `{ "flightTrackMode": "LAUNCH_BOOST" | "SUCTION_WALL_CLIMB" | "INVERTED_RELEASE" |
"BALLISTIC_ROLL_KICKFLIP" | "RETRO_HOVER_SLAM" | "HIGH_SPEED_VENTURI" |
"CRYOGENIC_ICE_YAW", "targetSpeedMps": number, "targetPitchDeg": number, "targetRollDeg":
number, "targetYawRateDegS": number, "targetAltitudeM": number, "suctionFanPowerKw":
number, "skirtTargetGapMm": number, "rearWingMode": "STOWED" | "MAX_DOWNFORCE" |
"AIRBRAKE_90", "cgtArmState": boolean, "reasoningTrace": string }`

2. TIER 2 — 250 Hz COUPLED TRACK-FLIGHT CONVEX MPC / SCvx CONTROLLER:
   - Over a 1.5-second receding horizon (`N = 30` steps of `50 ms`), solve the quadratic
trajectory tracking problem minimizing tracking error on position `p`, velocity `v`,
attitude quaternion error `q_err`, and angular velocity `omega`, penalized by nitrogen mass
consumption `w_gas * sum(m_dot_k)`:
     `min_{F_des, M_des} sum_{i=0..N} [ ||p_i - p_ref||^2_Qp + ||v_i - v_ref||^2_Qv +
||q_err,i||^2_Qq + ||omega_i - omega_ref||^2_Qw + ||u_i||^2_R ]`
   - Include specialized closed-loop state machines for:
     * **McGregor 170-Deg Wall Climb & Drop**: Maintain `F_suction = 46.1 kN` (`4,700 kgf`)
up the 21 m ramp to `pitch = 170 deg`; cut fans on drop command; level chassis attitude to
`[pitch=0, roll=0]` during free fall; trigger 3.3g (`76.1 kN`) hover-slam burn at `z = 6.5
m` using the 4 wheel-well lift nozzles + pitched rear nozzles to touch down at `v_z > -1.2
m/s`.
     * **Convoy Kickflip (360-Deg Axial Barrel Roll)**: At ramp liftoff (`v = 38 m/s`,
`z_apogee = 7.1 m`), fire all 4 wheel-well lift nozzles for vertical clearance while
pulsing left wheel-well nozzles (`CGT-W1, CGT-W3`) at `18.39 kN` total for `0.625 s`
(`+16.08 rad/s^2` roll acceleration), then fire right wheel-well nozzles (`CGT-W2, CGT-W4`)
at `18.39 kN` for `0.625 s` to arrest roll rate to `0.0 rad/s` at exactly `phi = 360.0 deg`
prior to touchdown.

3. TIER 3 — 1,000 Hz CONTROL ALLOCATION MATRIX & BLOWDOWN COMPENSATOR:
   - Map the desired 6-DOF wrench `W_des = [F_x, F_y, F_z, M_x, M_y, M_z]^T` across
available actuators in strict priority order:
     1) Tire longitudinal/lateral forces (RWD torque vectoring + 4-wheel independent
brakes) scaled by current normal load `F_N`,
     2) Active aerodynamic surfaces (front 80-deg wing, rear 90-deg Optimus wing) scaled by
dynamic pressure `0.5 * rho * v^2`,
     3) Underbody suction fans (`0..80 kW`),
     4) 10 SpaceX Cold-Gas Thrusters via constrained non-negative least-squares pseudo-
inverse of the 6x10 nozzle moment-arm matrix `B_cgt`, dynamically dividing by real-time
blowdown pressure `P_0(t)` to compensate for adiabatic pressure decay.
```

Stage 5: Ten Interactive 3D Adventure Proving Grounds & Automated Mission State Machines

Feed Stage 5 to Antigravity AI to build all 10 interactive 3D adventure environments, terrain shaders, obstacle convoys, and automated test trails:

```
[STAGE 5 PROMPT — EXECUTE IN ANTIGRAVITY AI]
Create the 10 playable and autonomously testable 3D Adventure Proving Ground environments
in `apps/web-simulator/src/adventures/` with an interactive Adventure Selector, procedural
3D terrain/ramp geometry, atmospheric/surface physics modifiers, camera choreography
(Cockpit Yoke View, Chase Drone View, Underbody Suction X-Ray View, Thermal/Shockwave Plume
View), and automated pass/fail telemetry verification:

1. `ADV-01` — **SpaceX McGregor 170° Vertical Wall & Inverted Ceiling Drop (Waco, TX)**:
```

   - 3D Scene: Nighttime McGregor launch complex with floodlights, Raptor test stand static fire in background, and a 21-meter-tall curved steel-and-concrete parabolic wall reaching a 170-degree inverted overhang.
   - Automated Trial: Autonomous launch -> 4,700 kgf suction lock -> climb to 21 m @ 170 deg pitch -> zero-speed inverted pause -> fan release & mid-air pitch/roll leveling -> 3.3g retro-thruster arrest at 6.5 m -> soft pad touchdown.

2. `ADV-02` — **Tesla Convoy Kickflip & 360° Mid-Air Barrel Roll (McGregor Pad, TX)**:
   - 3D Scene: Launch strip featuring a parked/moving 5-vehicle Tesla convoy (Tesla Semi, Cybertruck, Cybercab, Model Y, Model 3) flanked by takeoff and landing ramps.
   - Automated Trial: 0-60 mph in 0.81 s launch with Mach diamond rear plume -> ramp liftoff to 7.1 m altitude -> 360-degree axial roll over the convoy via asymmetric wheel-well CGT pulses -> level landing.

3. `ADV-03` — **Snake River Canyon 500-Meter Supersonic Ballistic Leap (Twin Falls, ID)**:
   - 3D Scene: 500-meter canyon gorge with a 28-degree launch ramp on the south rim, 35 km/h turbulent canyon crosswinds, and a recovery landing zone on the north rim.
   - Automated Trial: 2G fan-suction launch + full 80 kN rear CGT burn -> 4.64g acceleration to 238 mph takeoff -> 94 m apogee ballistic flight with active aero & CGT crosswind rejection -> north-rim retro-burn touchdown.

4. `ADV-04` — **Nürburgring Nordschleife Sub-5-Minute Autonomous Time Attack (Germany)**:
   - 3D Scene: High-speed Nordschleife sector sequence featuring Flugplatz crest, Karussell 18-degree banked concrete bowl, and Döttinger Höhe straight.
   - Automated Trial: Crest liftoff suppression via 4,700 kgf suction pulse -> automatic 25 mm skirt lift through Karussell banking -> 230+ mph straight -> 90-degree Optimus airbrake + 3.15g braking into chicane.

5. `ADV-05` — **Bonneville Salt Flats Mach 0.5 (383 MPH) High-Speed Aero Run (Utah)**:
   - 3D Scene: 15-kilometer crystalline salt flat corridor (`mu = 0.65`) with mirage heat-haze shader.
   - Automated Trial: FOD screen protection below 180 mph -> automatic transition at 200 mph to Open-Skirt High-Speed Venturi Mode + low-drag wing stowage -> sustained 600L COPV rear thruster burn pushing top speed past 330-368+ mph.

6. `ADV-06` — **Pikes Peak 14,115-Foot Thin-Air Alpine Ascent (Colorado)**:
   - 3D Scene: High-altitude switchback mountain road (`P_ambient = 60.1 kPa`, `-38%` air density) with sheer cliff drop-offs.
   - Automated Trial: Demonstrates 38% reduction in fan vacuum DeltaP alongside +10.5% increase in cold-gas nozzle specific impulse; autonomous hairpin apex rotation via nose yaw CGTs + RWD torque vectoring.

7. `ADV-07` — **Monaco Urban Tunnel Corkscrew & Yacht Helipad Hop (Monte Carlo)**:
   - 3D Scene: Curved concrete tunnel transitioning onto the harbor waterfront with a moving superyacht helipad 45 meters offshore.
   - Automated Trial: Continuous 112 km/h inverted ceiling corkscrew along the tunnel roof using 2G suction + centrifugal load -> exit portal -> precision 2.8-second ballistic CGT hop onto the heaving yacht pad (`<0.15 m` error).

8. `ADV-08` — **Moab Hell's Revenge Sandstone Crevasse Multi-Hop (Utah)**:
   - 3D Scene: Red slickrock canyon trail interrupted by a 16-meter-wide sheer rock crevasse.
   - Automated Trial: Suspension lifts +35 mm and retracts nylon-bristle skirts 100% into sills -> VLA occupancy scanner maps far ledge -> synchronized rear + wheel-well CGT launch vaulting the 16.4 m chasm.

9. `ADV-09` — **Arctic Sea-Ice Sheet -40°C Cryogenic Zero-Friction Slalom (Svalbard)**:
   - 3D Scene: Glare polar ice sheet (`mu = 0.12`, `T_ambient = -40 C`) with high-speed slalom gates.

```
   - Automated Trial: Octovalve thermal loop heats Optimus cryo-valves while internal COPV
gas drops to `-128.4 C` during blowdown; nose lateral yaw thrusters and rear gimbaled CGTs
execute 145 km/h zero-countersteer slalom.

10. `ADV-10` — **Giga Texas Autonomous Obstacle Gauntlet & Falcon Pad Hover-Slam (Austin,
TX)**:
    - 3D Scene: Giga Texas proving track with dynamic cyber-barriers and a circular SpaceX
landing pad.
    - Automated Trial: High-speed obstacle blockage at 18 m -> 14 ms VLA detection ->
emergency 4.6 m vertical CGT hop over barrier -> mid-air yaw turn -> closed-loop Falcon-9-
style hover-slam onto the center bullseye (`<0.10 m` error).
```

Stage 6: Chief Engineer Track Mode HMI, Industry Sim Exporters & Verification Suite

Feed Stage 6 to Antigravity AI to complete the cockpit UI, industry simulation exporters, and automated test suite:

```
[STAGE 6 PROMPT — EXECUTE IN ANTIGRAVITY AI]
Build the Chief Engineer Telemetry HUD, the 4 external simulator export generators, and the
automated verification test suite:

1. CHIEF ENGINEER TRACK MODE & FLIGHT HMI (`apps/web-
simulator/src/ui/ChiefEngineerHUD.tsx`):
   - Replicate the exact October 15, 2026 Tesla Roadster Track Screen (`Thruster · 5,500
PSI` bottom bar, `Dynamics` menu for Front Wing, Rear Wing, Vent Hood, Barge Boards, Aero
Load, and `Max Grip` Fan Mode):
     * Trim Switcher: `Performance ($50k)` | `Thrust ($250k)` | `Rocket (Invite Only - 1
Seat)`
     * Live COPV Blowdown Gauge: Pressure (`0..5,500..10,000 PSI`), Gas Mass (`0..250 kg`),
Cryo Temp (`+20°C to -130°C`), Compressor Recharge Toggle.
     * Live Ground-Effect Suction Gauge: Plenum Vacuum (`0..13.5 kPa`), Downforce
(`0..4,700 kgf`), Skirt Gap (`mm`), Fan Power (`0..80 kW`), Venturi Gate Status.
     * 10-Thruster Polar Vector Rose & 6-DOF G-Force Sphere (`Longitudinal g`, `Lateral g`,
`Vertical g`).
     * 1,048-Part Interactive CAD Tree & Exploded-View Slider (`0%..100%`) with X-Ray
Chassis & Thermal Plume view toggles.
     * VLA Reasoning Live Terminal displaying the 25 Hz JSON tactical thoughts and 1 kHz
MPC control allocation residuals.

2. INDUSTRY SIMULATION EXPORT BRIDGES (`exporters/`):
   - Generate downloadable, ready-to-run configuration bundles for:
     * `exporters/assetto_corsa/`: Complete `car.ini`, `aero.ini`, `drivetrain.ini`,
`suspensions.ini`, `tires.ini`, and CSP Extended Physics `script.lua` implementing the 80
kN blowdown thruster on Spacebar/Boost paddle and 4,700 kgf zero-speed fan LUT.
     * `exporters/beamng_drive/`: Complete `roadster26.jbeam` node-beam chassis definition
+ `lua/controller/roadsterSpaceXPackage.lua` implementing 10 thruster nodes and surface-
normal suction attraction for inverted loop driving.
     * `exporters/matlab_simulink/`: `init_roadster_2026_params.m` +
`run_copv_blowdown_mpc.m` script for MATLAB Vehicle Dynamics Blockset.
     * `exporters/unreal_engine_5/`: `ARoadsterChaosVehiclePawn.h/.cpp` + Niagara
`NS_ColdGasMachDiamond` parameter schema using `AddForceAtLocation` at all 10 thruster
sockets and 4 fan plenums.

3. AUTOMATED VERIFICATION & PHYSICS AUDIT SUITE (`tests/verify_roadster_physics.spec.ts`):
   - Write and execute 25 unit and integration tests verifying:
     * Exact 1,048 part count across `SYS-01`..`SYS-10` with zero missing Part IDs (`RD26-
0001`..`RD26-1048`).
```

```
     * Adiabatic blowdown from 379.2 bar (5,500 psi) yields `-51°C` at 50% gas mass and
`-105°C` at 25% gas mass within 1% tolerance.
     * 4 rear nozzles at 379.2 bar produce `>80 kN` thrust with `m_dot = 135..151 kg/s`.
     * 4 suction fans at `80 kW` and `3.5 mm` skirt gap produce `>=4,700 kgf` (`46.1 kN`)
downforce, holding the 2,350 kg car inverted at 170 degrees pitch.
     * 0-60 mph sprint times match `<1.60 s` (Performance), `0.81-0.89 s` (Thrust), and
`0.55-0.59 s` (Rocket).
     * All 10 Adventure state machines complete from start to finish with zero crashes or
NaN state divergences.
```

Stage 7: Ready-to-Compile Simulation Code Templates (Assetto Corsa, BeamNG.drive, MATLAB & Unreal Engine 5.5)

Feed Stage 7 to Antigravity AI to populate the exact physics configuration files and native C++/Lua/MATLAB scripts inside the `exporters/` directory:

```
[STAGE 7 PROMPT — EXECUTE IN ANTIGRAVITY AI]
Generate the following four production-ready physics simulation bridge modules inside
`exporters/`:

1. ASSETTO CORSA EXTENDED PHYSICS (`exporters/assetto_corsa/data/car.ini`, `aero.ini`,
`script.lua`):
   - In `car.ini`: Set `TOTALMASS=2350` (Rocket wet mass) or `1700` (Performance dry),
`INERTIA=1.98, 1.12, 4.55`, `CG_LOCATION=0.46`, `CG_HEIGHT=0.42`.
   - In `aero.ini`: Define `[WING_0]` (Front 5-Surface Active Wing, `CD=0.045`, `CL=0.85`,
`AOA=0..80`), `[WING_1]` (Rear Optimus Active Wing, `CD=0.065..0.95`, `CL=1.45`,
`AOA=0..90` airbrake yielding 700 kgf @ 160 mph), and `[WING_2]` (Underbody Suction Plenum
LUT generating `46,100 N` [4,700 kgf] downforce at `0 km/h` when `Max Grip` fan mode is
active).
   - In `script.lua` (CSP Extended Physics):
     ```lua
     local copv_pressure_bar = 379.2 -- 5,500 psi nominal track readout
     local gas_mass_kg = 240.0       -- 600L COPV real-gas N2/air charge
     local V_eff = 565.0             -- Sea-level under-expanded effective exhaust speed
(m/s)
     function script.update(dt)
       local car = ac.getCarState(1)
       if car.extraA and gas_mass_kg > 5.0 then -- Spacebar / Boost Paddle active
         local m_dot = 144.8 * (copv_pressure_bar / 379.2)
         local thrust_N = m_dot * V_eff
         gas_mass_kg = math.max(0.0, gas_mass_kg - m_dot * dt)
         local mass_ratio = math.max(0.05, gas_mass_kg / 240.0)
         copv_pressure_bar = 379.2 * math.pow(mass_ratio, 1.4)
         ac.addForce(vec3(0, 0, thrust_N), true, vec3(0, 0.38, -1.95), true)
       end
       if car.extraB then -- Underbody 4x20 kW Suction Fan System active
         local fan_downforce_N = (car.speedKmh < 320.0) and 46100.0 or 18500.0
         ac.addForce(vec3(0, -fan_downforce_N, 0), true, vec3(0, 0.15, 0.0), true)
       end
     end
     ```

2. BEAMNG.DRIVE SOFT-BODY & 10-NOZZLE LUA CONTROLLER
(`exporters/beamng_drive/lua/roadsterSpaceX.lua`):
   - Bind 10 directional `thrusters` node pairs in `roadster26.jbeam` (`cgt_r1..r4` at rear
diffuser, `cgt_w1..w4` at wheel arches, `cgt_n1..n2` at front nose fenders).
   - Apply surface-normal suction force `F_suction = -46100 * surface_normal_vec` across
the 16 floor-pan JBeam nodes (`f1ll..f4rr`) so the vehicle can drive up the 21-meter
```

```
McGregor curved wall to 170 degrees inverted pitch without detaching.
   - Implement the closed-loop 360-degree Kickflip and 6.5-meter 3.3g hover-slam state
machine inside `updateGFX(dt)`.

3. UNREAL ENGINE 5.5 CHAOS VEHICLES C++ COMPONENT
(`exporters/unreal_engine_5/RoadsterSpaceXComponent.cpp`):
   - Subclass `UActorComponent` attached to `AWheeledVehiclePawn`.
   - In `TickComponent(float DeltaTime, ...)`, query
`UPrimitiveComponent::AddForceAtLocation` for all 10 cold-gas thruster sockets
(`Socket_CGT_R1`..`Socket_CGT_N2`) and the 4 underbody fan plenums
(`Socket_Fan_1`..`Socket_Fan_4`).
   - Drive the `UNiagaraComponent` parameter `User.MachDiamondIntensity` and
`User.PlumeMachNumber` (`2.18f` to `4.70f`) in real time from `InstantThrustN`.

4. MATLAB / SIMULINK 6-DOF & SCvx MPC VERIFICATION SCRIPT
(`exporters/matlab_simulink/run_roadster_mpc.m`):
   - Simulate the coupled 18-state ODE (`ode45` / fixed-step `ode4` at 1,000 Hz) across all
10 Adventure Proving Grounds and export CSV/JSON telemetry traces for regression comparison
against the WebGPU simulator.
```

Appendix: Quick-Reference Configuration Matrix for Assetto Corsa, BeamNG & Unreal Engine 5

When executing Stage 6 in Antigravity AI or porting the generated parameters directly into external desktop simulation tools, the following verified baseline constants govern the exported physics files (CellCog; EVXL; Kingy AI):

Parameter Symbol / Key	Assetto Corsa (`*.ini` / CSP Lua)	BeamNG.drive (`*.jbeam` / Lua)	Unreal Engine 5.5 (Chaos Vehicles C++)
Vehicle Mass (Dry / Rocket Wet)	`TOTALMASS=1700` (Perf) / `2350` (Rocket)	Node weights summed to `1700..2350 kg`	`Mass = 2350.0f` (dynamic via `SetMassOverrideInKg`)
Center of Gravity Height & Bias	`CG_LOCATION=0.46` (46% F / 54% R), `CG_HEIGHT=0.42`	CG node at `[0.0, 0.12, 0.42] m`	`CenterOfMassOffset = FVector(-12.0f, 0.0f, 42.0f)`
RWD Twin-Motor Drivetrain	`TYPE=RWD`, `FINAL_RATIO=8.40`, `MAX_POWER=1000..1100 kW`	2 independent rear `electricMotor` devices, `gearRatio: 8.4`	Rear-only torque distribution + custom `UChaosVehicleWheel` torque vectoring
4 × 20 kW Suction Fan Downforce	`aero.ini` Wing LUT with `CL_GAIN` active at `0 km/h` = `46100 N`	Lua `obj:applyForceVector` along negative surface normal = `46100 N`	`AddForceAtLocation(-UpVector * 4610000.0f, PlenumCenter)` (in cm/g/s units)
10 SpaceX Cold-Gas Thrusters	CSP `ac.addForce` at 10 socket offsets with blowdown decay	10 JBeam `thrusters` pairs driven by `electrics.values.cgt_0..9`	10 `USceneComponent` sockets calling `AddForceAtLocation` at 1,000 Hz